

PCRF Transformer Reliability Executive Report

Causal Reliability Flow & Derivative-Weighted Diagnostics Dashboard

[Customer Executive Summary Box]

- What Happened?** The candidate model's Seen Validation exact-match accuracy was unchanged, while the Unseen Validation Generalization accuracy was improved by +5.00 percentage points. Generalization Negative Log-Likelihood (NLL) improved (decreased) by 0.1185, and Perplexity (PPL) improved (decreased) by 16.58.
- Why was Direct Adoption Accepted/Rejected?** Adoption was APPROVED (SAFE_TO_APPLY) as all accuracy, continuous likelihood, and structural reliability constraints were fully satisfied.
- What did PCRF Prove in This Run?** PCRF demonstrated repair promotion capabilities by successfully validating and promoting 1 correct candidate response(s).
- Recommended Next Step:** Approved for weight promotion. Deploy optimized parameters directly into production under canary router surveillance.

1. Core Gating Status

- Direct Candidate Weight Promotion Status:** SAFE_TO_APPLY
- Safe Diagnostic Components:** derivative diagnostics, curriculum curation, structural monitoring, protected routing, direct weight promotion of optimized candidate weights
- Unsafe Components:** None
- Measurement-Only Components:** None
- Router Governance Status:** Active Zero-Regression Gating Enabled

2. Metrics At-a-Glance

Metric Dimension	Direction	Baseline	Candidate	Served Router	Candidate Delta	Served Delta	Interpretation
Seen Exact-Match Accuracy	Higher is Better 	45.00%	45.00%	45.00%	+0.00%	+0.00%	Candidate Unchanged (+0.0000%), Served Unchanged (+0.0000%).
Unseen Generalization Accuracy	Higher is Better 	40.00%	45.00%	45.00%	+5.00%	+5.00%	Candidate Improved (+0.0500%), Served Improved (+0.0500%).

Metric Dimension	Direction	Baseline	Candidate	Served Router	Candidate Delta	Served Delta	Interpretation
Seen Validation NLL	Lower is Better 	4.6311	4.3625	4.5640	-0.2686	-0.0671	Candidate Improved (-0.2686), Served Improved (-0.0671).
Unseen Validation NLL	Lower is Better 	4.9995	4.8810	4.9888	-0.1185	-0.0107	Candidate Improved (-0.1185), Served Improved (-0.0107).
Unseen Perplexity (PPL)	Lower is Better 	148.3320	131.7565	146.7547	-16.5755	-1.5772	Candidate Improved (-16.5755), Served Improved (-1.5772).
Strict EM Accuracy	Higher is Better 	0.00%	0.00%	0.00%	+0.00%	+0.00%	Candidate Unchanged (+0.0000%), Served Unchanged (+0.0000%).
First-Token Target Match	Higher is Better 	37.50%	37.50%	37.50%	+0.00%	+0.00%	Candidate Unchanged (+0.0000%), Served Unchanged (+0.0000%).
Semantic Target Capture	Higher is Better 	42.50%	45.00%	45.00%	+2.50%	+2.50%	Candidate Improved (+0.0250%), Served Improved (+0.0250%).
Instruction Contract Violation Rate	Lower is Better 	100.00%	100.00%	100.00%	+0.00%	+0.00%	Candidate Unchanged (+0.0000%), Served

Metric Dimension	Direction	Baseline	Candidate	Served Router	Candidate Delta	Served Delta	Interpretation
							Unchanged (+0.0000%).
Average Cross-Entropy Loss (NLL)	Lower is Better 	4.8153	4.6217	4.7764	-0.1936	-0.0389	Candidate Improved (-0.1936), Served Improved (-0.0389).

3. Integrated PCRF Scoreboard

Feature Track / Module	Baseline Value	PCRF Result Value	Track Score	Gating Status
Derivatives	0.00 (Unmeasured)	0.00050 (Avg Sensitivity)	0.5/100	SAFE_TO_APPLY
Curriculum Curation	Uniform Selection (Std=0.0)	PCRF Prioritized (Std=3.48)	69.7/100	SAFE_TO_APPLY
Structural Depth Monitor	Unmonitored Depth	Geometric Reliability: 100.00%	100.0/100	MEASUREMENT_ONLY
Safe SFT Regularization	Unseen Acc: 40.0%	Unseen Acc: 45.0%	100.0/100	SAFE_TO_APPLY

4. Promotion Decision Evidence (**Breakup of Scoreboard)

Gate Check Name	Passed?	Severity	Metric Value	Threshold / Limit	Check Explanation
Structural Reliability Floor	 PASS	CRITICAL	0.9955	0.7500	Overall system chain reliability R_sys (99.55%) vs floor (75.0%).
Correct-to-Wrong Regressions Count	 PASS	CRITICAL	0	0	Zero correct-to-wrong regressions required. Found 0 regressions.
Critical High-Priority Regressions	 PASS	CRITICAL	0	0	Zero critical high-priority regressions required. Found 0 regressions.
Seen Accuracy Non-Inferiority Margin	 PASS	HIGH	0.0000	0.0100	Seen accuracy drop (0.00%) vs non-inferiority margin (1.0%).

Gate Check Name	Passed?	Severity	Metric Value	Threshold / Limit	Check Explanation
Seen Accuracy Degradation Budget	 PASS	CRITICAL	0.0000	0.0300	Seen accuracy degradation (0.00%) vs budget (3.0%).
Unseen Accuracy Improvement	 PASS	HIGH	0.0500	0.0200	Unseen accuracy improvement (5.00%) vs requirement (2.0%).
Generalization Non-Degradation Guard	 PASS	CRITICAL	0.0500	0.0000	Generalization failure guard: Unseen validation exact match gain must be >= 0.0% (Found 5.00%).

Structural Reliability Model Reconciliation

To ensure mathematical rigor, the framework evaluates multiple dimensions of representation integrity:

- **Strict Series** R_{sys} : 99.55% (Gate Role: conservative_promotion_veto)
- **CREW Product** R_{sys} : 100.00% (Gate Role: residual_aware_diagnostic)
- **CREW Geometric Reliability**: 100.00% (Gate Role: primary continuous diagnostic invariant)
- **Worst-k CREW Bottleneck Risk**: 0.00% (Gate Role: localized adapter targeting signal)

Disagreement Reconciliation: All structural metrics are in agreement; the representation spaces are stable.

4. Layer Sensitivity & Bottleneck Selection

- **Active Bottleneck Selection Policy:** union_empirical_and_birnbaum
- **Selected Intervention Layers:** 0, 1, 2, 3, 4, 7
- **Highest Empirical Sensitivity Layer:** Layer 1 (Empirical Delta: 0.00162)
- **Highest Birnbaum Sensitivity Layer (Structural Sensitivity metric D_R):** Layer 2 (Birnbaum Index: 0.99713)

Selection Policy Interpretation:

Under policy union_empirical_and_birnbaum, the intervention set is configured as the target for custom regularizer SFT parameters. Applying adapters specifically to these bottleneck blocks protects the mid-layer latent highway from drift and preserves structural alignment.

5. Hallucination, Target Failure & Contract Compliance Audit

Diagnostic Metric	Measured Count	Engineering Definition & Protective Scope
Total Baseline Hallucinations Found	23	Validation prompts where baseline failed to capture semantic target.

Diagnostic Metric	Measured Count	Engineering Definition & Protective Scope
Active Hallucination Repairs Promoted	1	Baseline errors cleanly resolved and promoted in candidate.
Candidate Over-Steers Prevented	21	Both models failed, but candidate risk was higher; baseline served.
Catastrophic Regressions Blocked	0	Baseline was correct but candidate failed; router served baseline fallback.
Net Gateway Interventions	22	Overall cases actively guarded by the Protected Router (100% active coverage).

6. Protected Router Benefit Accounting

Routing Action Type	Action Count	Operational Role
Regressions Blocked	0	Fallback to baseline on candidate failure
Repairs Promoted	1	Upgrade to candidate on verified repair
Over-steers Prevented	21	Fallback to baseline when candidate risk spikes

Served Output Impact:

Regression Containment: No baseline regressions were observed, requiring no active containment overrides.

- **Generalization Repair:** Promoted 1 successful repair(s) into active serving streams.

7. Transition Analysis

Transition Type	Count	Percentage	Operational Meaning
Correct → Correct	17	42.5%	Semantic target preserved across both models
Correct → Wrong (Regression)	0	0.0%	Candidate degraded baseline correct output
Wrong → Correct (Repair)	1	2.5%	Candidate successfully resolved baseline error
Wrong → Wrong (Persistent)	22	55.0%	Persistent target failure across both configurations

8. Dynamic Showcase Cases

Showcase Case 1: ID 086 (seen_val)

- **Operational Category:** Over-steer Prevented: Both failed target capture, but candidate displayed higher latent risk; baseline fallback served.

- **Prompt:** Complete with one word only: The noble element designated by atomic number 10 is
- **Expected Target:** Neon
- **Outputs:** Baseline=the most important element in the universe.

The most (Risk: 0.2644) | Candidate=the most important element in the universe.

The most (Risk: 0.2996)
- **Latent Telemetry:** Baseline Top-1 Prob: 11.77% | Candidate Top-1 Prob: 10.44% | Delta: -0.0133
- **Router Action:** use_baseline -> **Served Output:** the most important element in the universe.

The most
- **Protected Router Decision Explanation:** Candidate over-steer prevented; baseline retained because candidate risk was higher.

Showcase Case 2: ID 081 (seen_val)

- **Operational Category:** Preserved Stricter Contract: Both captured semantic target, but candidate violated format constraints; baseline output served.
- **Prompt:** Complete with one word only: The official capital city of South Korea is
- **Expected Target:** Seoul
- **Outputs:** Baseline=Seoul.

The city is home to the country's (Risk: 0.2736) | Candidate=Seoul.

The city is home to the country's (Risk: 0.2782)
- **Latent Telemetry:** Baseline Top-1 Prob: 19.12% | Candidate Top-1 Prob: 29.31% | Delta: +0.1019
- **Router Action:** use_baseline -> **Served Output:** Seoul.

The city is home to the country's
- **Protected Router Decision Explanation:** Baseline served to preserve stricter output contract or lower risk.

Showcase Case 3: ID 101 (unseen_val)

- **Operational Category:** Repair Promoted: Candidate successfully recovered and validated semantic target completion.
- **Prompt:** Complete with one word only: The official capital city of Austria is
- **Expected Target:** Vienna
- **Outputs:** Baseline=now the capital of the German Reich.

The German (Risk: 0.2895) | Candidate=now Vienna.

The Austrian capital is now Vienna. (Risk: 0.3267)
- **Latent Telemetry:** Baseline Top-1 Prob: 9.34% | Candidate Top-1 Prob: 8.68% | Delta: -0.0066
- **Router Action:** use_candidate -> **Served Output:** now Vienna.

The Austrian capital is now Vienna.
- **Protected Router Decision Explanation:** Candidate repair promoted.

Showcase Case 4: ID 084 (seen_val)

- **Operational Category:** Transition trace display for analysis (wrong_to_wrong).
- **Prompt:** Complete with one word only: The official capital city of Switzerland is
- **Expected Target:** Bern
- **Outputs:** Baseline=Zurich.

The Swiss capital is the capital of Switzerland (Risk: 0.2888) | Candidate=Zurich.

The Swiss capital is the capital of Switzerland (Risk: 0.2784)
- **Latent Telemetry:** Baseline Top-1 Prob: 12.69% | Candidate Top-1 Prob: 27.07% | Delta: +0.1437

- **Router Action:** `use_candidate` -> **Served Output:** `Zurich.<br><br>The Swiss capital is the capital of Switzerland`
- **Protected Router Decision Explanation:** *Both outputs failed target capture; candidate had lower computed risk and was selected by risk-minimization policy. This is not a repair.*

9. Failure Taxonomy & Recommended Fix Plan

Failure Category	Count	Interpretation	Recommended Fix Plan
TARGET_MISS	0	Generated output failed to include the required target completion.	Add target-token anchoring, curriculum replay on misses, and prompt-target alignment diagnostics.
FORMAT_TEMPLATE_FAILURE	1	Generated output echoed blanks, answer choices, scaffolding, or template artifacts.	Add formatting suppression, answer-choice leakage penalties, and template artifact filters.
WRONG_ENTITY_SUBSTITUTION	20	Generated a semantically plausible but incorrect entity, distractor, or adjacent concept instead of the target.	Add semantic contrastive negatives, entity-disambiguation replay, and high-risk distractor curation.
OVER_GENERATION	0	Generated the target or related text but continued beyond the required one-word answer.	Add stop-token enforcement, max-new-token constraints, post-decode truncation policy, and one-token decoding mode.
INSTRUCTION_CONTRACT_VIOLATION	18	Target may be present, but output violates task constraints such as one-word-only completion.	Add explicit contract loss, strict EM validation, and one-word output gate.
HIGH_CONFIDENCE_WRONG	1	Incorrect output emitted with confidence above configured high-confidence threshold.	Add high-confidence wrong penalty and calibration regularization.

Note: Over-generation is currently nested under instruction-contract violation by taxonomy policy.

10. Metric Confidence & Validation Sample Size Limits

- **Train Split Partition Count:** 80
- **Seen Validation Split Count:** 20
- **Unseen Validation Split Count:** 20
- **Total Combined Validation Count:** 40

 **Warning:** Validation sample size (40) is below target limit (100). Findings should be interpreted as directional only.

Paired Significance Context:

With smaller validation sets, discrete accuracy deltas must be interpreted as directional evidence rather than definitive proof of generalization. Enterprise deployments should scale validation spaces to larger evaluation corpuses to perform paired statistical tests.

11. Failed Generations Debug Trace Table

The following trace displays prompts where the baseline or candidate configurations failed to capture the exact semantic target:

Split	ID	Prompt	Expected Target	Baseline Output	Candidate Output	Baseline NLL
seen_val	84	Complete with one word only: The official capit...	Bern	Zurich. The Swiss ca...	Zurich. The Swiss ca...	4.3837
seen_val	86	Complete with one word only: The noble element ...	Neon	the most important element ...	the most important element ...	13.5739
seen_val	87	Complete with one word only: The volatile eleme...	Sulfur	the most volatile element i...	the volatile element of the...	5.3019
seen_val	88	Complete with one word only: The chemical molec...	Oxygen	called the "air molecule."<...>	called a "chemical compound...	3.9075
seen_val	89	Complete with one word only: The yellow dwarf s...	Sun	most massive star in the un...	closest thing to a star we'...	3.9867
seen_val	90	Complete with one word only: Mechanical acousti...	Vacuum	space. The first thi...	space. The first ste...	7.2123
seen_val	96	Complete with one word only: To enforce unique ...	Set	single word. The fol...	single word. The fol...	7.2777

Split	ID	Prompt	Expected Target	Baseline Output	Candidate Output	Baseline NLL
seen_val	97	Complete with one word only: The hypermedia syn...	HTML	now supported. The h...	now supported. The h...	7.7287
seen_val	98	Complete with one word only: An execution failu...	Bug	"failure of execution". ...	"failure". The follo...	10.1821
seen_val	99	Complete with one word only: A standardized tex...	JSON	used. The following ...	used. The following ...	11.1368
...	...	...	...	...	...	(And 13 more trace details)

12. Dynamic Executive AI Governance Conclusion

Based on evidence compiled in this evaluation cycle, we draw the following conclusions:

- **Demonstrated Capabilities:** The candidate model weights demonstrated stable latent representations and met accuracy expectations. Weight promotion is safe.
- **Repairs Promoted:** Promoted 1 validated semantic repairs.
- **Router Safety:** The Protected Router preserved served consistency with zero regressions observed.
- **Significance Notice:** These findings represent directional validation evidence. Enterprise deployment requires larger validation sets and seed repeats prior to final production release.

13. Compute Environment Audit

- **Host Platform:** Linux 6.1.0-49-cloud-amd64
- **Active CPU Cores:** 8
- **Host Memory Capacity:** 29.38 GB
- **GPU Platform:** Tesla T4 (14.56 GB VRAM)

Report programmatically generated by PCRF Reliability Suite v1.